

Final Project Proposal: Campus Map Optimizer

By: Landon Schultz

For my final project, I want to make a route optimizer for the Stillwater campus. This would allow for one to find the most optimal route from one location to another under different sets of conditions. To make the project more realistic, I want to include multiple transportation types such as walking, biking, scooters, and cars. Different methods of travel would have different constraints on where they are allowed to go. For example, cars would only be able to use roads, while walking, bikes, and scooters could use both roads and sidewalks. I would also assign different values to the path segments depending on things like distance, estimated speed, or travel type restrictions. This would allow the route chosen by the program to change depending on how the user is traveling, which makes the problem more interesting and more realistic. For how numerical methods would be implemented, the campus will be treated like a set of nodes and weighted connections, allowing me to mathematically analyze the optimal routes. Matrix or linear algebra based representations can be used to store and organize the relationships between locations and paths. Interpolation could also be used to estimate values between known points if needed, such as approximating travel costs or smoothing estimated movement behavior across portions of campus. Root finding may also be useful if I want to determine when one type of route becomes better than another, such as when a scooter route becomes faster than walking over a certain distance. Along with this, I also plan on including some kind of comparison or error analysis by testing how changes in speed assumptions or weighting conditions affect the resulting route. In short, this is just going to be a more complex and in depth version of solving a maze.

Project Goals:

- Create a python program that models the Stillwater campus as a network of connected locations.
- Represent roads and sidewalks as different path types between nodes.
- Allow the user to choose a start location, end location, and mode of transportation.
- Assign different weights to path segments based on distance, estimated speed, and travel restrictions.
- Determine the most efficient route based on a selected optimization condition such as shortest distance or fastest travel time.
- Use interpolation where needed to estimate route values between known points.
- Display the final route in a clear and readable format.
- Create a visual plot of the campus map and highlight the chosen path.
- Compare how different travel modes and weighting conditions affect the resulting route.

Measurable Requirements:

- The program shall model the Stillwater campus as a route network containing at least 20 connected nodes with connecting path segments. These path segments will represent either roads or sidewalks.
- The program shall correctly compute and display the optimal route between any two valid campus locations for at least 4 transportation types: walking, biking, scooter, and car. The output must include both the selected path and its total distance or travel time.
- The program shall enforce travel restrictions with 100% accuracy across at least 10 test cases. For example, cars must never be assigned to sidewalks, while walking, biking, and scooters may use both sidewalks and roads.
- The program shall implement at least 1 interpolation-based feature and produce estimated route values within 5% error of directly computed values. This may be used for approximating travel costs or values between known path points.
- The program shall demonstrate route correctness using at least 3 test cases with known expected results, where the computed route cost is within 5% of the expected optimal value. These test cases will be used to verify that the route optimizer is functioning properly.